fragmentation, sporangiospores, conidia etc., Meiosis is zygotic. Example : *Peziza*

Recently, with the advent of molecular methods myxomycetes and oomycetes were reclassified and treated under chromista.

The salient features of some of the classes – Oomycetes, Zygomycetes, Ascomycetes, Basidiomycetes and Form class Deuteromycetes are discussed below.

Oomycetes

Coenocytic mycelium is present. The cell wall is made up of Glucan and Cellulose. Zoospore with one whiplash and one tinsel flagellum is present. Sexual reproduction is oogamous. Example: *Albugo.*

Zygomycetes

- Most of the species are saprophytic and live on decaying plant and animal matter in the soil. Some lead parasitic life (Example: *Entomophthora* on housefly).
- Bread mold fungi (Example: *Mucor, Rhizopus*) and coprophilous fungi (Fungi growing on dung Example: *Pilobolus*) belong to this group (Figure 1.20).

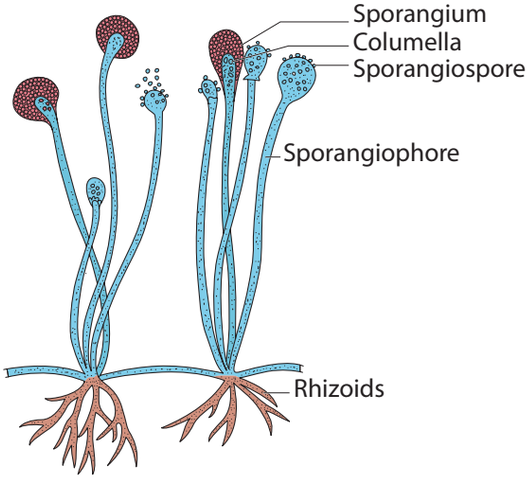
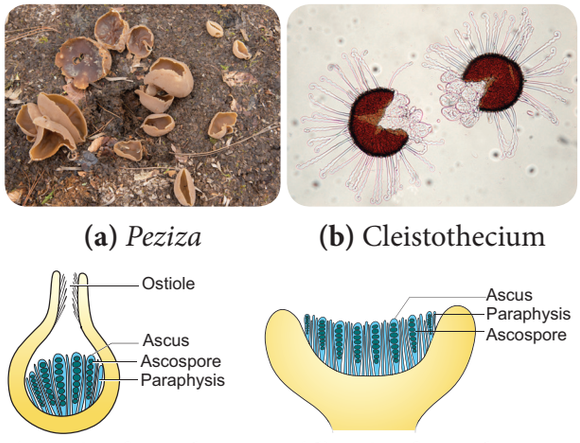


Figure 1.20: Zygomycetes – *Rhizopus*

- The mycelium is branched and coenocytic.
- Asexual reproduction by means of spores produced in sporangia.
- Sexual reproduction is by the fusion of the gametangia which results in thick walled zygospore. It remains dormant for long periods. The zygospore undergoes meiosis and produce spores.

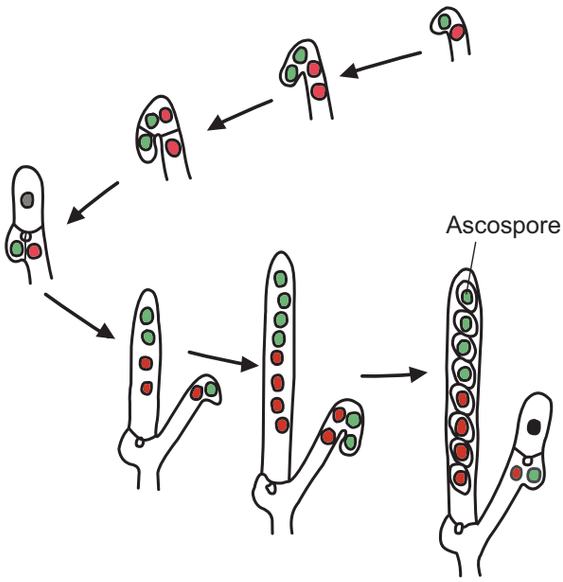
Ascomycetes

- Ascomycetes include a wide range of fungi such as yeasts, powdery mildews, cup fungi, morels and so on (Figure 1.21).



(a) *Peziza* **(b)** Cleistothecium

(c) V.S. of Perithecium **(d)** V.S. of Apothecium



(e) Steps involved in the development of Ascus

Figure 1.21: Structure and reproduction in Ascomycetes

- Although majority of the species live in terrestrial environment, some live in aquatic environments both fresh water and marine.
- The mycelium is well developed, branched with simple septum.
- Majority of them are saprophytes but few parasites are also known (Powdery mildew – *Erysiphe*).

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